

WALL ROVER

PROJECT REPORT (Phase II)

*Submitted in partial fulfilment of the requirements for the award of
B. Tech Degree in Electronics and Communication Engineering
from the APJ Abdul Kalam Technological University*

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We declare that this report titled **WALL ROVER** in partial fulfilment for the award of the degree of Bachelor of Technology in **Electronics and Communication Engineering** of APJ Abdul Kalam Technological University, is a record of original work carried out by us and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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ABSTRACT

This paper presents the design, development, and implementation of Wall Rover equipped with AI-based surface detection and adaptive suction control. The robot employs a distributed multi-microcontroller architecture featuring an ESP32-CAM running a YOLOv11 object detection model for real-time classification of wall surfaces into seven categories: glass, concrete, wood, metal, plastic, brick, and drywall. An STM32F103C8T6 serves as the central controller, managing an RS220S 2300KV brushless motor-driven propeller fan for adhesion, while an additional ESP32 module coordinates wheel locomotion via an L298N motor driver and hosts a web-based remote control interface. The system integrates HC-SR04 ultrasonic sensing for obstacle detection, HC-05 Bluetooth for real-time distance monitoring, and a robust power distribution network centered around a 7.4V 3000mAh LiPo battery. This integrated approach enables autonomous surface adaptation, with suction power dynamically adjusting from 60% on smooth glass to 95% on porous concrete based on real-time AI classification.

The mechanical and control systems are designed to achieve reliable vertical traversal across diverse surface types. Four N20 DC motors provide independent wheel control for forward, backward, and turning movements, while the ducted propeller fan generates negative pressure within a sealed chamber to secure the robot to vertical surfaces. The gap height between the flexible sealing lip and wall surface is maintained at 1-3mm to maximize adhesion efficiency. Software development within the Arduino IDE ecosystem enables seamless communication between the three microcontrollers, with the STM32 processing AI commands from the ESP32-CAM and coordinating motor responses. Safety protocols including automatic emergency stop upon obstacle detection within 5cm, low-voltage cutoff at 6.0V, and continuous battery monitoring ensure reliable operation during extended climbing missions. The web interface hosted on the ESP32 provides manual override capability, allowing operators to intervene when necessary while monitoring real-time telemetry data.

The adaptive adhesion principle validated here, combined with edge-based AI inference on resource-constrained hardware, represents a significant advancement toward practical deployment of climbing robots in real-world unstructured environments.

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LIST OF SYMBOLS AND ABBREVIATIONS

YOLO	You Only Look Once
GPIO	General Purpose Register Input/Output
BLDC	Brushless Direct Current
AI	Artificial Intelligence
BT	Bluetooth
ESC	Electronic Speed Controller
IDE	Integrated Development Environment
LiPo	Lithium Polymer
N20	Nano 20mm
PWM	Pulse Width Modulation
RPM	Revolution Per Minute
CFD	Computational Fluid Dynamics

CHAPTER 1

INTRODUCTION

The development of wall-climbing robots has emerged as a transformative solution for accessing vertical and inverted surfaces that are hazardous or inaccessible to humans. These robots find extensive applications in infrastructure inspection, high-rise building maintenance, bridge monitoring, and disaster response scenarios. This project presents an intelligent wall-climbing robot that integrates artificial intelligence for real-time surface detection with an adaptive suction-based adhesion mechanism. The system employs an ESP32-CAM running YOLOv11 for surface classification, an STM32F103C8T6 for centralized control, and an ESP32 module for web-based remote operation. A propeller-based suction system driven by an RS2205 brushless motor provides variable adhesion force based on detected surface type, while four N20 DC motors enable locomotion across diverse wall materials including glass, concrete, wood, metal, plastic, brick, and drywall.

1.1. Objective

The field of service robotics is rapidly expanding, with an increasing demand for machines capable of operating in complex and hazardous environments. A significant frontier within this domain is vertical locomotion, where robots are designed to navigate and perform tasks on walls, facades, and other vertical structures. The automation of tasks such as visual inspection, maintenance, and cleaning on surfaces like skyscrapers, bridges, and historical monuments offers tremendous potential for improving efficiency, reducing costs, and enhancing human safety by eliminating the need for dangerous manual work. The development of reliable climbing robots is, therefore, a critical step towards the future of automated infrastructure management.

1.2. Adhesion Systems for Climbing Robots: A Review

Adhesion mechanisms represent the most critical subsystem in wall rover design, with several approaches developed over decades of research. Permanent magnet adhesion offers high force with minimal power consumption but is strictly limited to ferromagnetic surfaces like steel. Traditional vacuum suction systems generate negative pressure using pumps, providing strong adhesion on smooth surfaces but suffering from high power consumption and poor performance on porous materials. Propeller-based

negative pressure adhesion, the approach adopted in this project, offers a lightweight alternative by using high-speed rotating blades within a ducted housing to create suction without mechanical pumps. Bio-inspired adhesion systems mimicking geckos or spiders remain largely experimental with challenges in scalability and durability. Hybrid systems combining multiple principles increase versatility but add mechanical complexity. For this project, propeller-based adhesion was selected for its optimal balance of lightweight construction, low power consumption relative to generated force, and adaptability to diverse surface types through speed control.

1.3. The Need for Adaptability in Unstructured Environments

Most existing climbing robots are designed for flat surfaces, limiting their utility on real-world structures filled with irregularities, protrusions, and varying textures. A robot's ability to maintain stable adhesion and traction while navigating these obstacles is paramount. This project addresses the adaptability challenge through a hybrid wheeled-legged configuration. The robot employs underactuated legs that passively conform to the surface profile, ensuring all wheels maintain ground contact. This mechanical adaptability ensures continuous traction and stability, allowing the robot to traverse obstacles that would immobilize a rigid-bodied platform.

1.4. Potential Applications and Scope

The applications for a versatile vertical locomotion robot are vast. They can be deployed for structural health monitoring of bridges and dams, routine inspection of high-rise building facades, cleaning and painting of large industrial structures, and even archaeological surveys of historical monuments. This project focuses on the development of a robot capable of operating on concrete and stone surfaces, with a design emphasis on mechanical robustness, power efficiency, and autonomous functionality.

CHAPTER 2

LITERATURE REVIEW AND TECHNICAL BACKGROUND

2.1 Literature review

The field of vertical locomotion robotics has seen significant advancements in recent years, with researchers exploring various adhesion and locomotion mechanisms. Orozco-Magdaleno et al. (2025) pioneered the hybrid wheeled-legged configuration with vortex adhesion, demonstrating remarkable adaptability to surface irregularities through underactuated mechanisms. Their work established the foundation for mechanical adaptability in climbing robots. In parallel, Kim and Cho (2020) developed an EDF-based climbing robot, providing crucial insights into vortex adhesion dynamics and gap optimization. Watanabe and Matsuno (2018) contributed to the field with their negative pressure adhesion and crawler mechanism, offering solutions for stable vertical movement on various surfaces.

The integration of sensing and control systems has been equally crucial. Wang et al. (2020) demonstrated the effectiveness of sensor fusion algorithms in maintaining robot stability, while Bochkovski et al. (2020) advanced real-time object detection capabilities through YOLOv5, enabling sophisticated environmental perception. Lu et al. (2021) further enhanced this paradigm by incorporating machine vision feedback for surface-adaptive climbing, and Tanaka and Suzuki (2022) implemented SLAM-based navigation for precise inspection capabilities. Kim and Lee (2019) contributed with their impeller-type adhesion mechanism, offering alternative approaches to vortex-based adhesion systems.

Table.No.2.1: Literature Review

Name of the Author	Name of Journal / Work	Summary / Contribution
Adriano Cavalcanti,	Advanced Robotics	Hybrid climbing robot with four wheeled-legs and EDF for vertical adhesion; adaptable

Name of the Author	Name of Journal / Work	Summary / Contribution
Pablo Ramirez, Fabiano Silva		design for façade inspection using vortex suction; reference for leg design, adhesion, and sensor integration
Wang Feng, Liu Jiang, Chen Yu	Advanced Robotics	Two-wheeled self-balancing robot using MPU6050 (gyro + accelerometer) with PID control for stability; sensor fusion + PID ensures upright posture and supports multi-directional remote navigation
Watanabe KazuFumitoshio, Matsuno	Advanced Robotics	Development of a wall climbing robot using negative pressure vacuum adhesion and crawler mechanism; addresses suction failure, angle limitations, and load distribution; reference for adhesion force
Bochkovskiy A., Wang C.-Y., Liao H.-Y.M.	arXiv Preprint: Real-Time Object Detection with YOLOv5 for Autonomous Navigation	YOLOv5 real-time object detection using CNNs for fast and accurate classification + localization; applied in project for surface analysis via HD Camera to adjust EDF suction; technical insights for material-aware adhesion
Lu J., Han L., Zhou M.	Advanced Robotics	Climbing robot with machine vision; detects surface color, texture, and patterns to classify wall types; camera-based feedback enables real-time adjustment of climbing behavior; supports CNN-driven AI module validating adaptive EDF suction control

Name of the Author	Name of Journal / Work	Summary / Contribution
Jonghoek Kim, Hyeoncheol Lee	Journal of Intelligent & Robotic Systems	Wall-climbing robot using impeller-type adhesion mechanism; EDFs for suction-based climbing optimizing speed, airflow, and placement;

2.2 System Introduction

The proposed system represents an integration of mechanical innovation and intelligent control, building upon existing research while introducing novel contributions. The robot employs a four-legged wheeled configuration with underactuated mechanisms, similar to Orozco-Magdaleno's approach, but enhances it with real-time material identification capabilities. The core system comprises three integrated subsystems: a vortex adhesion mechanism using EDF technology, an adaptable wheeled-leg locomotion system, and an intelligent vision-based control system. This integration enables the robot to not only navigate vertical surfaces but also dynamically adjust its adhesion parameters based on surface characteristics.

The system's intelligence layer incorporates advances from multiple research streams. The material identification module draws inspiration from Bochkovskiy's YOLOv5 architecture and Lu's surface adaptation work, while the stability control system implements sensor fusion principles similar to Wang's approach. The EDF control system builds upon Kim's and Cho's research on vortex adhesion optimization. This comprehensive integration creates a robust platform capable of autonomous operation on diverse vertical surfaces, from smooth glass to rough concrete, while maintaining optimal adhesion force and power efficiency.

2.3 Adhesion Technologies for Vertical Surfaces

Adhesion technologies for climbing robots have evolved significantly, with each approach offering distinct advantages for specific applications. Pneumatic adhesion systems, particularly vortex-based mechanisms as demonstrated by Kim and Cho

(2020) and Orozco-Magdaleno et al. (2025), provide excellent performance on rough and irregular surfaces without direct contact. These systems generate adhesion through high-speed rotors creating low-pressure zones, making them ideal for stone and concrete surfaces. Watanabe and Matsuno (2018) showed that negative pressure adhesion with proper sealing can achieve high reliability on smoother surfaces.

Alternative adhesion methods include mechanical systems using claws or spines, suitable for porous surfaces but limited in speed and maneuverability. Electro-adhesion has emerged as a promising technology for smooth surfaces, utilizing electrostatic forces for attachment. Kim and Lee (2019) demonstrated that impeller-type adhesion mechanisms offer compact solutions with quick engagement and disengagement capabilities. The selection of adhesion technology depends critically on surface characteristics, with vortex systems particularly advantageous for the rough, non-ferromagnetic surfaces targeted in this project due to their robustness and non-contact operation.

2.4. Locomotion Mechanisms : Wheel, Legs and Tracks

Locomotion mechanisms for climbing robots present distinct trade-offs between speed, adaptability, and complexity. Wheeled systems, as employed in Orozco-Magdaleno's design, offer high speed and control simplicity but limited obstacle negotiation capabilities. Tracked systems, similar to Watanabe and Matsuno's approach, provide better weight distribution and traction but suffer from higher power consumption and mechanical complexity. Legged systems excel in obstacle negotiation but typically involve complex control algorithms and lower movement speed.

The hybrid wheeled-legged configuration adopted in this project represents an optimal compromise, combining the efficiency of wheels with the adaptability of legs. This approach allows the robot to maintain high mobility while adapting to surface irregularities through underactuated mechanisms. Each leg incorporates two rotational joints with tendon-driven actuation, enabling passive adaptation to surface contours while wheels maintain continuous propulsion. This design philosophy aligns with Orozco-Magdaleno's findings that hybrid systems can achieve both efficiency and adaptability in vertical environments.

2.5. Vortex Adhesion : Principles and Aerodynamics

Vortex adhesion operates on the principle of generating a low-pressure zone through high-speed rotational airflow, creating suction force without physical contact with the surface. The fundamental physics follows Bernoulli's principle, where high-velocity airflow within the EDF shroud creates reduced pressure relative to the ambient environment. Kim and Cho (2020) established that the adhesion force (F_s) can be expressed as $F_s = A_{in} \times P_a$, where A_{in} represents the effective pressure area and P_a is the pressure differential.

The aerodynamic efficiency of vortex systems depends critically on several parameters. Orozco-Magdaleno et al. (2025) demonstrated through CFD analysis that the gap distance (h) between the shroud and surface significantly affects adhesion performance. Optimal performance typically occurs at intermediate gap distances where airflow restriction balances pressure differential generation. The shroud geometry, rotor speed, and surface characteristics all influence the system's efficiency. Kim and Lee's (2019) research on impeller systems further confirmed that careful aerodynamic design can substantially enhance adhesion force while minimizing power consumption.

2.6. Machine Vision for Material Identification

Machine vision has revolutionized robotic perception, enabling sophisticated environmental understanding and adaptive behavior. Bochkovskiy et al. (2020) demonstrated the remarkable capabilities of YOLOv11 for real-time object detection, achieving high accuracy with computational efficiency suitable for embedded systems. This advancement provides the foundation for real-time material identification in dynamic environments. Lu et al. (2021) further advanced this field by implementing surface-adaptive climbing using vision feedback, showing that visual cues can effectively guide robotic adhesion control.

Material identification for adhesion control requires specialized approaches beyond conventional object detection. Texture analysis, surface pattern recognition, and color-based classification can distinguish between materials like concrete, glass, brick, and metal. The integration of these visual features with machine learning algorithms enables robust material classification under varying lighting conditions. Tanaka and Suzuki's (2022) work on SLAM-based navigation further demonstrates how visual information

can be fused with other sensor data to create comprehensive environmental models for autonomous operation.

2.7. Integration of Sensing and Actuators

The effective integration of sensing and actuation systems is crucial for autonomous robotic performance. Wang et al. (2020) demonstrated that sensor fusion algorithms combining IMU data with other sensory inputs can significantly enhance stability and control precision. Their approach to self-balancing through multi-sensor integration provides valuable insights for maintaining stable adhesion in climbing robots. The proposed system extends this principle by fusing visual information from the material identification system with EDF control parameters.

Modern autonomous systems require hierarchical control architectures that process sensory information at multiple levels. Lu et al. (2021) showed that vision feedback can be effectively integrated with low-level control loops for real-time adaptation. Similarly, Tanaka and Suzuki's (2022) SLAM-based navigation demonstrates how high-level planning can be combined with reactive control. In the proposed system, this integration enables dynamic adjustment of adhesion force based on identified surface materials, optimizing power consumption while maintaining safety margins. The system implements closed-loop control where visual perception directly influences actuation parameters, creating an adaptive and efficient climbing platform.

CHAPTER 3

SYSTEM DESIGN AND METHODOLOGY

3.1. System Design

The wall rover features a distributed multi-processor architecture designed for optimal task allocation and real-time performance. The STM32F103C8T6 serves as the central controller, managing critical functions including motor control, sensor data acquisition, safety monitoring, and inter-processor communication. It generates precise PWM signals for the ESC to control the BLDC motor speed, coordinates wheel movements through the L298N motor driver, and continuously monitors battery voltage and ultrasonic distance readings. The STM32 operates at 72MHz, providing sufficient processing power for real-time control loops while maintaining low power consumption for extended battery life.

The ESP32-CAM functions as the dedicated vision processing unit, capturing video at usable frame rates and running the YOLOv11 object detection model for simultaneous surface classification and crack detection. The camera module is positioned at the robot's front with an optimal viewing angle, and the onboard LED flash provides illumination for low-light conditions. The ESP32-CAM communicates detected surface types and crack locations to the STM32 via UART at 115200 baud, enabling real-time adaptive responses. A second ESP32 Module hosts a web-based control interface and manages the L298N motor driver for wheel movements, receiving commands either from the web interface or forwarding them from the STM32 based on operational mode.

The mechanical chassis is constructed from lightweight yet rigid material, housing all electronic components while maintaining a low center of gravity for stable wall traversal. Four N20 DC motors with integrated gearboxes provide independent wheel drive, each mounted with high-traction wheels suitable for vertical surface adhesion. The suction system consists of an RS2205 2300KV brushless motor driving a custom-designed ducted propeller fan within a sealed chamber. A flexible sealing lip around the chamber perimeter maintains optimal 1-3mm gap from the wall surface, maximizing negative pressure generation. The L298N motor driver controls all four

wheel motors independently, with PWM inputs enabling smooth speed regulation and direction control.

The power distribution system centers around a 7.4V 3000mAh LiPo battery connected through a main power distribution board. XL4015 DC-DC step-down converters regulate voltage to 5V for sensors, ESP32-CAM, and HC-05 module, and 3.3V for the STM32. The 30A BLDC ESC receives direct battery power for the high-current suction motor. An HC-SR04 ultrasonic sensor mounted at the robot's front provides continuous obstacle distance measurement, with data transmitted to a smartphone via the HC-05 Bluetooth module for remote monitoring. The complete system design prioritizes modularity, allowing individual component testing and replacement while maintaining robust integration for reliable wall climbing operation.

3.2. Control System Architecture and Block Diagram Analysis

The control system architecture of the wall-climbing robot is a distributed, hierarchical embedded design with the STM32F103C8T6 microcontroller serving as the central processing unit. It coordinates all critical operations: receiving surface classification results from the ESP32-CAM, computing the required suction thrust, controlling the propeller fan speed via the 30A ESC and RS2205 BLDC motor, managing wheel locomotion through the L298N motor driver and N20 DC motors, and processing obstacle data from the HC-SR04 ultrasonic sensor. The system supports both autonomous mode (surface-adaptive suction and obstacle avoidance) and manual override mode via the HC-05 Bluetooth module, which transmits real-time commands and sensor data (e.g., obstacle distance) to a smartphone app.

As shown in the block diagram, the ESP32-CAM captures a live video feed of the wall surface and runs the YOLOv11 object detection model to classify the material (glass, concrete, wood, metal, plastic, etc.). The detected surface type is immediately sent to the STM32, which calculates the optimal suction power level based on a pre-defined mapping table and generates the corresponding PWM signal to the ESC for precise fan speed adjustment. The HC-SR04 ultrasonic sensor measures forward distance to obstacles and sends this data to the STM32. The STM32 then forwards the obstacle distance information via the HC-05 Bluetooth module to the user's phone for real-time monitoring and safety alerts. In parallel, the STM32 directly controls the L298N driver

to adjust the speed and direction of the N20 DC motors for wheel-based movement, ensuring stable rolling once suction holds the robot to the wall.

The architecture is designed for low-latency response and fault tolerance: the STM32 acts as the master controller, receiving inputs from vision (ESP32-CAM), distance sensors (HC-SR04), and user commands (HC-05), while issuing outputs to propulsion (ESC/BLDC for suction) and locomotion (L298N/N20 motors). An additional ESP32 module is included for potential future WiFi-based features (e.g., web dashboard or live streaming). The entire system is powered by a 7.4V LiPo battery regulated through the XL4015 DC-DC converter and power distribution board, ensuring stable 3.3V/5V supplies for the microcontrollers and sensor. This block diagram clearly illustrates the flow of data from perception (vision & sensor) through decision-making (STM32) to actuation (motors & fan), forming a robust closed-loop control system for safe and adaptive wall climbing.

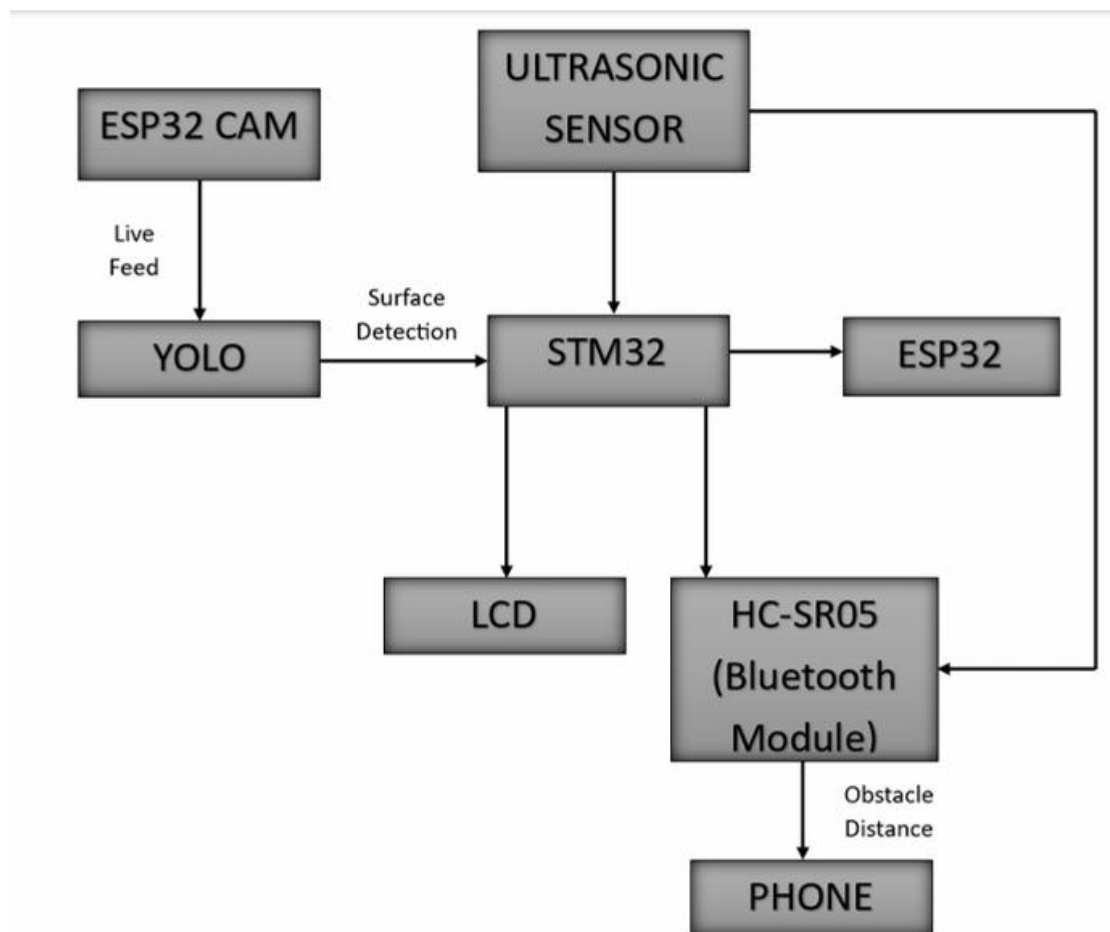


Fig.No.3.1: Block diagram of control part

3.2.1. Control logic and actuation

A pre-calibrated look-up table maps the identified material to an optimal propeller PWM value. For example:

IF material == "Rough Concrete" THEN set Propeller PWM = 85%

IF material == "Smooth Glass" THEN set Propeller PWM = 60%

This logic is based on the principle that rough surfaces allow more air leakage, requiring higher suction power, while smooth surfaces can maintain a seal with less power. This command is sent from main STM32 controller to ESP32, which then adjusts the signal to the propeller's ESC.

3.3. Locomotion Architecture and Block Diagram

The locomotion architecture of the wall rover is a hybrid thrust-assisted wheeled system centered on the ESP32 as the primary actuator controller for both wheel movement and suction generation. As shown in the block diagram, the ESP32 receives high-level commands and surface classification data from the STM32F103C8T6 (the central processor) and directly drives two separate output paths: one for wheeled locomotion and the other for vertical suction thrust. This distributed control allows the ESP32 to handle time-critical motor and ESC signals independently while the STM32 focuses on decision-making based on surface type (from ESP32-CAM + YOLOv11), obstacle distance (from HC-SR04 ultrasonic), and user inputs (via HC-05 Bluetooth).

The left path of the diagram shows the ESP32 interfacing with the L298N motor driver, which controls the speed and direction of the N20 DC motors. The L298N receives PWM signals for speed (ENA/ENB) and digital direction signals (IN1–IN4) from the ESP32, enabling forward, backward, left, right, and stop movements of the four wheels. This wheeled subsystem provides precise rolling and positioning on the wall surface once suction holds the robot in place. The right path illustrates the ESP32 driving the 30A ESC, which regulates power to the RS2205 2300KV BLDC motor and attached propeller fan. The ESC accepts PWM signals from the ESP32 to adjust fan speed dynamically, generating variable downward thrust that creates suction force for adhesion on different wall materials.

This architecture ensures real-time responsiveness: the STM32 calculates suction requirements based on surface type and sends instant commands to the ESP32, which translates them into precise PWM values for both the L298N (wheel control) and ESC (fan/suction control). The block diagram clearly separates the two propulsion mechanisms—wheeled locomotion for horizontal adjustment and thrust-based suction for vertical stability—while highlighting the ESP32’s role as the common bridge executing low-level actuator commands. The overall design supports both autonomous operation (surface-triggered suction and obstacle avoidance) and manual control (via Bluetooth), providing a balanced, modular, and efficient locomotion solution for reliable wall climbing. BT Car Controller-Arduino/ESP app is used to control the movement of the wheel and also suction power.

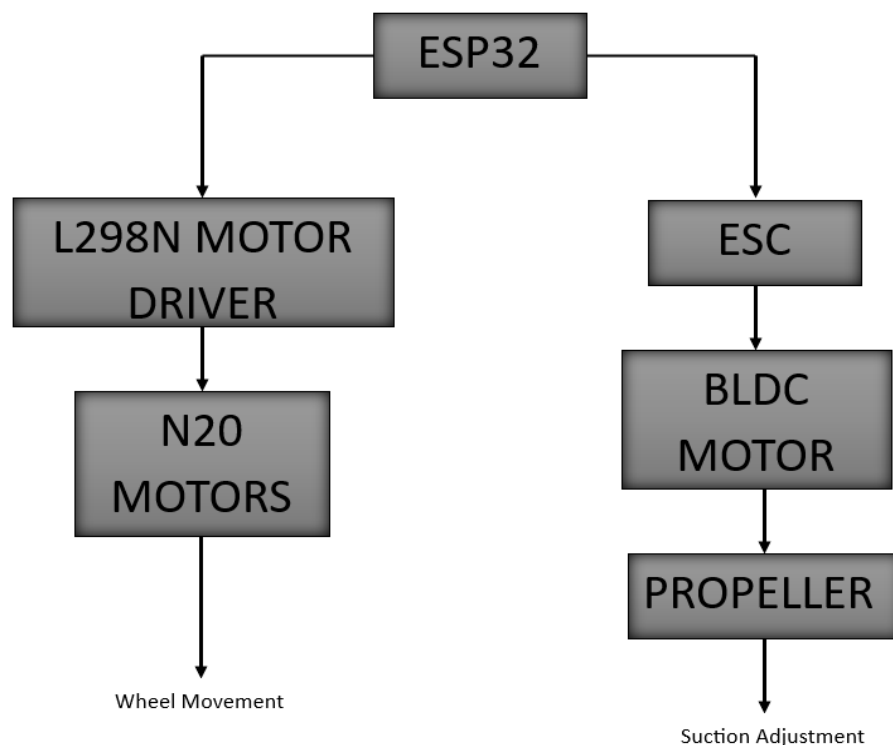


Fig.No.3.2: Block diagram of moving part

3.3.1. Vortex adhesion system

The adhesion force (F_s) generated by the propeller is derived from the pressure differential between the low-pressure zone inside the shroud (P_a) and the ambient air (P_b), acting over the effective area (P_b). The fundamental equations are:

$$F_s = A_{in} \times P_a \dots \dots \dots (3.1)$$

$$P_a = \frac{1}{2} \times v^2 \times P_b \dots \dots \dots (3.2)$$

where v is the airflow velocity. The effective area A_{in} is a function of the shroud radius (r_s) and the critical gap distance (h) between the shroud and the wall:

$$A_{in} = 2\pi h r_s \dots \dots \dots (3.3)$$

The adhesion force is highly dependent on the gap h . As the gap increases beyond an optimal point (e.g., >10 mm), airflow restriction decreases, leading to a drop in suction force. The propeller's high RPM is necessary to generate the required pressure differential to hold the robot's weight and payload against gravity.

3.4 Circuit Diagram

The circuit diagram of the AI-based wall rover is a well-organized, multi-module embedded system centered around the STM32F103C8T6 as the master microcontroller. The ESP32-CAM captures live video and runs the YOLOv11 model for surface classification (glass, concrete, wood, metal, plastic), transmitting the detected surface type via UART (TX/RX pins) to the STM32 for immediate suction power calculation. The STM32 then generates a PWM signal (on PA0 or similar TIM2 channel) to the 30A BLDC ESC, which powers the RS2205 2300KV brushless DC motor and propeller fan to adjust downward thrust and suction force dynamically based on surface type.

Wheel locomotion is controlled through the L298N motor driver, with the STM32 sending direction signals (IN1–IN4) and PWM signals (ENA/ENB) to drive the four N20 DC motors for forward, backward, left, right, and stop movements. Obstacle detection is handled by the HC-SR04 ultrasonic sensor (Trig on PB5, Echo on PB6), with distance data relayed via the HC-05 Bluetooth module (TX/RX on PA9/PA10) to a smartphone for real-time monitoring. Power is supplied by a 7.4V LiPo battery through a power distribution board and regulated by an XL4015 DC-DC converter to provide stable 5V and 3.3V rails to the STM32, ESP32 modules, sensors, and L298N logic. An additional ESP32 module supports future WiFi features or web-based control. The circuit is designed for low-latency response, modular debugging, and reliable operation on vertical surfaces.

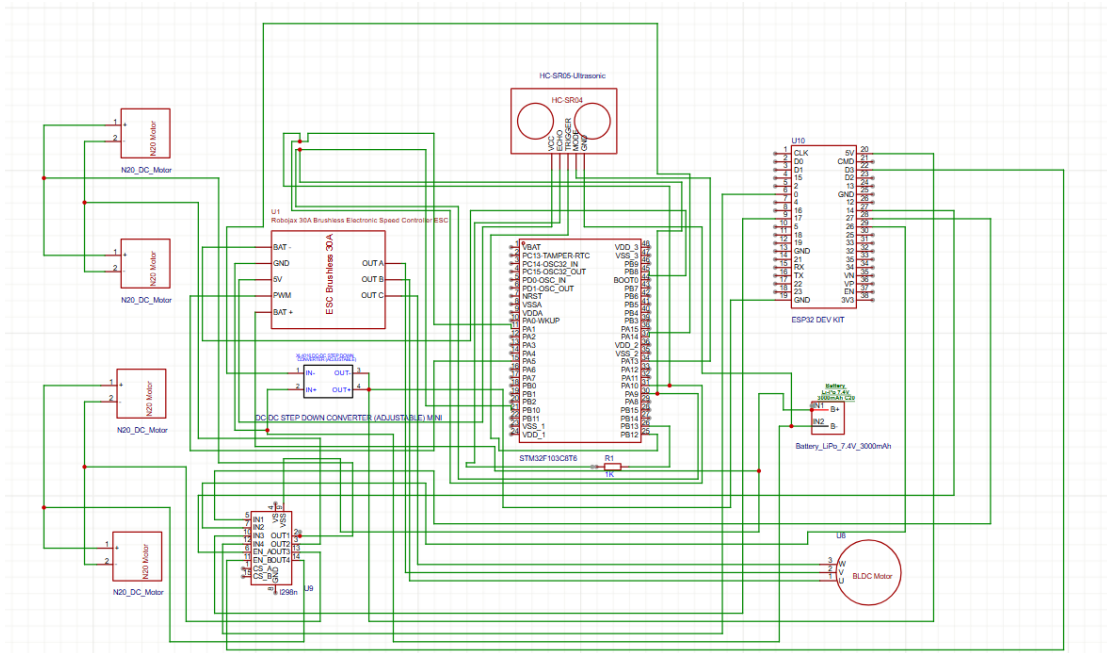


Fig.No.3.3: Circuit diagram

3.5. Components and System Architecture

The robot's architecture integrates computational, sensory, and actuation components to achieve intelligent vertical locomotion. The selection of components prioritizes processing capability for machine vision, precision for control, and power for adhesion and locomotion.

3.5.1 System architecture

1. Computational Hardware

The computational core uses a multi-processor architecture to distribute tasks effectively. The main control unit is an STM32F103C8T6 microcontroller, chosen for its real-time control capabilities, leg actuation, and wheel control. For the advanced task of material identification, an ESP32-CAM module is employed. This unit, with its integrated camera and dual-core processor, handles image capture and processing. A more powerful ESP32 microcontroller (Quad-core, 4 GB RAM) is included for controlling wheel movement.

2. Sensor Suite

Ultrasonic Sensor (HC-SR04): Used for coarse distance measurement to the wall, helping in initial positioning and detecting large surface projections.

3. Actuation System

The physical interaction with the environment is handled by a powerful and precise actuation system: Propeller with ESC: The primary adhesion mechanism. A brushless DC motor generates a vortex, and a 40A Electronic Speed Controller (ESC) allows for precise PWM-based speed control to modulate suction pressure. Wheels with DC Geared Motors: Provide locomotion. The rubber grip ensures traction on various vertical surfaces.

3.5.2 Components and specifications

1. STM32F103C8T6

The STM32F103C8T6, part of STMicroelectronics' STM32 family, is a cost-effective 32-bit ARM Cortex-M3 microcontroller running at up to 72 MHz. It features 64 KB flash memory, 20 KB SRAM, and extensive peripherals including ADCs, timers, UART, SPI, I2C, USB, and CAN interfaces. Ideal for embedded systems, IoT, and hobbyist projects like the Blue Pill board, it offers low power and robust performance.



Fig.No.3.4: STM32F103C8T6

2. ESP 32 CAM & ESP 32

The ESP32-CAM is a compact, low-cost module featuring an ESP32-S microcontroller with a dual-core processor, OV2640 camera, and Wi-Fi/Bluetooth connectivity. Ideal for IoT, surveillance, and robotics, it supports image capture, video streaming, and wireless communication, with GPIO pins



Fig.No.3.5: ESP 32



Fig.No.3.6: ESP 32 CAM

3. Propeller with casing

The propeller fan, housed within a specially designed casing or duct, serves as the critical component that converts the RS2205 brushless motor's rotational energy into the negative pressure (suction force) required for wall adhesion. When spun at high speed, the propeller accelerates air through the duct, creating a low-pressure region between the robot and the wall surface. This pressure differential generates the adhesion force that keeps the robot securely attached to vertical and inverted surfaces.

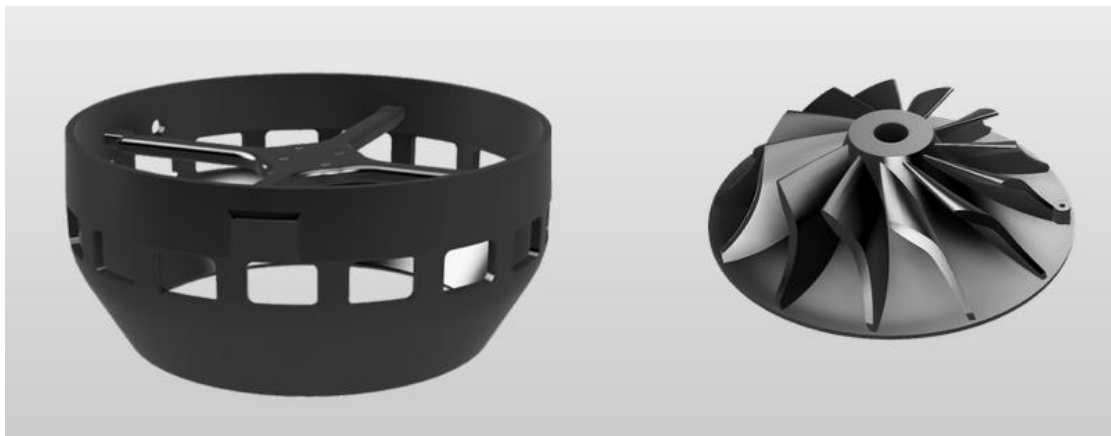


Fig.No.3.7: Propeller with casing

4. Ultrasonic Sensor

The HC-SR04 ultrasonic sensor is a low-cost, high-precision device for measuring distance. It uses ultrasonic waves to detect objects, offering a range of 2cm to 400cm. With simple interfacing via trigger and echo pins, it's ideal for robotics, obstacle detection, and IoT projects, providing reliable, non-contact measurements.

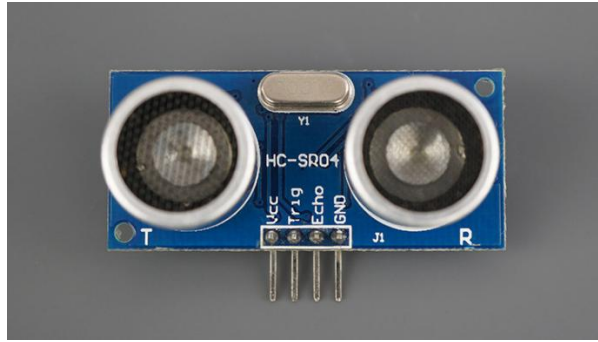


Fig.No.3.8: HC SR-04 Ultrasonic Sensor

5. HC-05 Bluetooth Module

The HC-05 Bluetooth module serves as a wireless communication bridge between the wall climbing robot and the user's smartphone. It continuously transmits real-time distance data from the ultrasonic sensor to a mobile app, allowing the operator to monitor obstacle proximity remotely. This provides critical safety feedback during manual control operation and helps in debugging sensor performance without needing a computer connection.

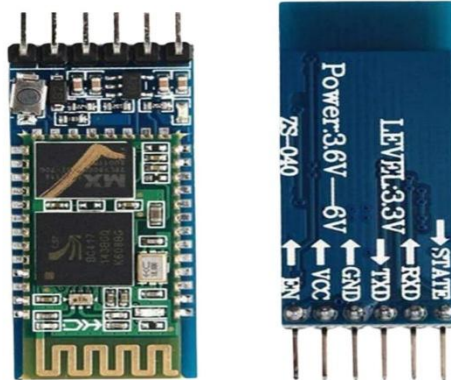


Fig.No.3.9: HC 05 Bluetooth module

6. XL4015 DC-DC Step Down Converter

The XL4015 DC-DC step-down converter serves as the primary voltage regulation module for the entire wall climbing robot system. It steps down the 7.4V from the LiPo battery to stable 5V and 3.3V required by all electronic components. This ensures clean, regulated power delivery to sensitive components like the STM32, ESP32-CAM, sensors, and Bluetooth module while isolating them from motor-induced voltage fluctuations.

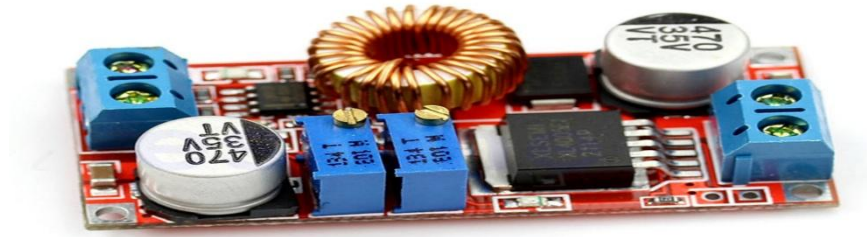


Fig.No.3.10: XL4015 DC-DC Step Down

7. 30A Brushless ESC

The 30A Brushless Electronic Speed Controller (ESC) serves as the precise power regulator for the RS2205 2300KV brushless motor that drives the propeller fan. It converts the STM32's low-voltage PWM signals into three-phase high-current power required to spin the motor at variable speeds. This enables real-time suction control based on surface type detected by the AI system, from gentle adhesion on glass to maximum grip on rough concrete.



Fig.No.3.11: Brushless ESC

8. L298N Motor Driver

The L298N motor driver serves as the dual H-bridge controller for all four N20 DC motors that drive the robot's wheels. It converts low-current STM32 control signals into high-current power needed to drive the motors forward or backward. This enables

precise speed and direction control for maneuvering the robot across walls, with individual wheel control for smooth turning and climbing.

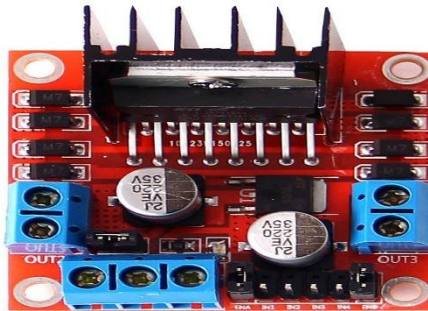


Fig.No.3.12: L298N Motor Driver

9. N20 DC Motor

The N20 DC micro motors serve as the primary locomotion system for the wall climbing robot, driving all four wheels to enable movement across horizontal and vertical surfaces. These compact gear motors provide the perfect balance of torque and speed needed for climbing while maintaining the robot's lightweight design. Each motor is independently controlled via the L298N driver, allowing precise differential steering for smooth navigation and wall-to-wall transitions.



Fig.No.3.13: N20 Motor

10. Brushless DC Motor (RS2205 2300KV)

The RS2205 2300KV brushless DC motor serves as the high-speed powerplant for the suction system, driving the propeller fan to create vacuum pressure for wall adhesion. It generates the necessary thrust force (up to 1.2kg with proper propeller) to keep the robot securely attached to vertical and even inverted surfaces. The motor's rapid response to PWM signals allows real-time suction adjustment based on surface type detected by the AI system.



Fig.No.3.14: Brushless DC Motor (RS2205 2300KV)

11. LiPo 7.4V 3000mAh C20 Battery

The 7.4V 3000mAh LiPo battery serves as the single power source for the entire wall climbing robot system, providing energy to all motors, controllers, sensors, and the AI processing unit. Its high discharge capability ensures sufficient current for simultaneous operation of four N20 wheel motors and the high-power RS2205 brushless suction motor. The battery's compact energy density allows extended operation time while maintaining the lightweight design essential for vertical wall adhesion.



Fig.No.3.15: LiPo Battery

IMPLEMENTATION AND DEVELOPMENT

4.1. Control System Design and Simulation

The control system architecture for the robot is designed as a hierarchical multi-loop system to ensure stable vertical locomotion and intelligent adhesion control.

- **High-Level Control (ESP32-CAM):** This unit acts as the brain of the operation. It runs the YOLOv11 object detection model to analyze video feed and classify the wall surface (e.g., glass, concrete). Based on the surface type, it calculates the required suction power and transmits this command to the STM32. This process is analogous to the visual localization and remote target acquisition used in teleoperation systems.
- **Mid-Level Control (STM32):** As the central coordinator, the STM32 receives the surface command from the ESP32-CAM and the desired wheel movements from the ESP32 (which hosts the website). It then calculates the precise control signals needed. For the propulsion system, this involves generating the appropriate PWM signals for the ESC to control the BLDC motor's speed, directly impacting suction strength. For locomotion, it computes the differential speed and direction commands for the four N20 DC motors via the L298N driver, similar to model-based control strategies used for accurate wheel motion.
- **Low-Level Control (Actuators):** This layer consists of the physical components that execute commands. The 30A ESC interprets the PWM signal from the STM32 to drive the RS2205 BLDC motor, which in turn spins the propeller fan to generate suction. Simultaneously, the L298N motor driver receives signals to independently control the speed and direction of each N20 motor, enabling the robot to move forward, backward, and turn.

The performance of your ducted propeller fan is critical. The adhesion force is not just about motor speed; it's highly dependent on the gap height between the robot's sealing mechanism and the wall surface. Computational Fluid Dynamics (CFD) software can be used to model the airflow within the propeller's duct and chamber. This helps optimize the fan blade design and the shape of the chamber to maximize the generated

negative pressure (vacuum) for different surface porosities, a technique validated in recent research on novel suction techniques.

4.2. AI Based Surface Recognition System Development

The AI-based surface recognition system of wall rover is built around the YOLOv11 object detection model, running on the ESP32-CAM as the dedicated vision processor. YOLO (You Only Look Once) is a state-of-the-art, single-stage detector that frames object detection as a regression problem, directly predicting bounding boxes and class probabilities from full images in a single evaluation . The choice of YOLOv11—the latest iteration in this renowned family—is strategic for your application: it offers an optimal balance between inference speed and detection accuracy, which is critical for real-time operation on resource-constrained hardware like the ESP32-CAM.

The development process, as outlined in various embedded vision tutorials, follows a structured pipeline. First, a custom dataset of wall surface images—glass, concrete, wood, metal, plastic, brick, drywall—is captured using the ESP32-CAM itself, ensuring that training data matches the operational conditions . These images are then uploaded to a cloud-based machine learning platform; Edge Impulse Studio, which streamlines the steps of data labeling, model training, and deployment . In the labeling phase, each image is annotated by drawing bounding boxes around the relevant surface areas and assigning the appropriate class label . This labeled dataset forms the foundation for training the YOLOv11 model to recognize the distinct visual features of each material type.

Equations to find accuracy & efficiency of YOLO training model:-

- (a) F1 Object Score:- The F1 object score is the harmonic mean of precision and recall for object instances.

$$F_{1,k} = \max_c \frac{2TP(c)}{2TP(c)+FP(c)+FN(c)} \dots \dots \dots (4.1)$$

where TP(c): True Positives, FP(c): False Positives, FN(c): False Negatives.

- (b) Mean average precision (MAP):- mAP is the macro-average of AP scores across all object classes.

$$mAP = \frac{1}{K} \sum_K AP \dots \dots \dots (4.2)$$

where, K represents number of object classes.

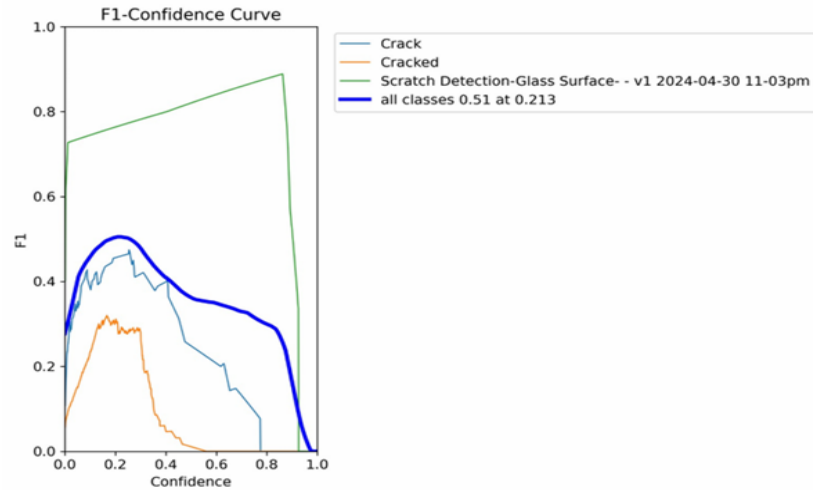


Fig.No.4.1: F1 Confidence curve

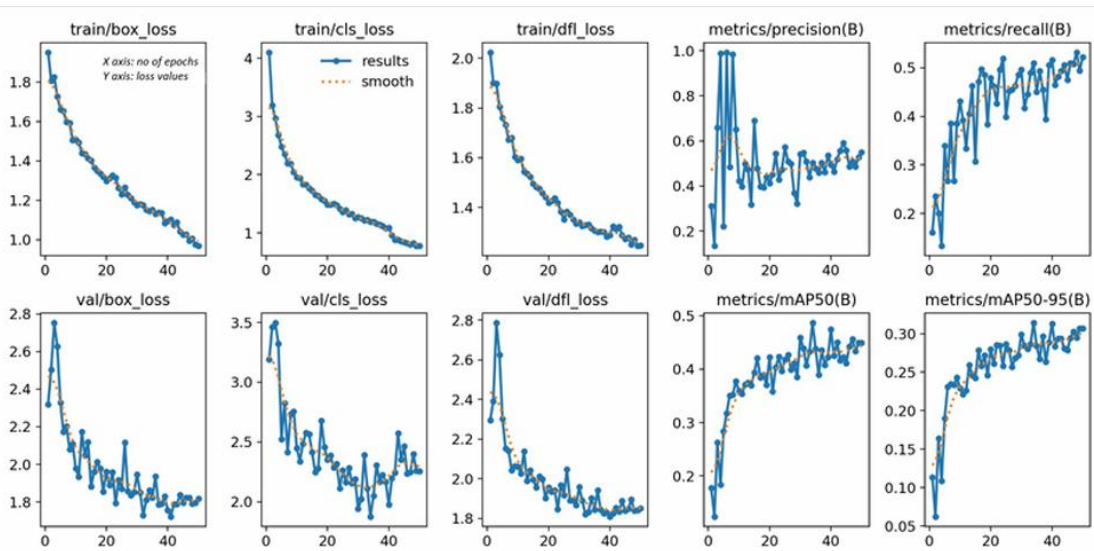


Fig.No.4.2: Working intrepretation of YOLO

4.3. Propeller and Suction Adaption Mechanism

The propeller and suction adaptation mechanism in wall rover operates as a closed-loop control system that dynamically adjusts adhesion force based on real-time surface analysis. When the ESP32-CAM identifies a surface type (glass, concrete, wood, etc.) using the YOLOv11 model, it sends this classification to the STM32, which consults a predefined surface-to-suction mapping table to determine the required propeller speed. The STM32 then generates a precise PWM signal (20-100kHz) to the 30A ESC, which converts this signal into three-phase power for the RS2205 2300KV brushless motor.

As the motor spins the ducted propeller fan at speeds ranging from 1000 to 2000 RPM, it creates negative pressure within the sealed chamber between the robot and wall surface. The gap height (typically 1-3mm) between the flexible sealing lip and the wall is critical, as smaller gaps exponentially increase suction force. This enables the robot to deliver gentle adhesion (60% power) on smooth glass to prevent surface damage, while providing maximum grip (95% power) on porous concrete or brick where air leakage is higher. The system continuously monitors ultrasonic distance and battery voltage as safety checks, ensuring the robot maintains optimal proximity to the wall and has sufficient power for sustained adhesion during vertical traversal.

$$P_a = \frac{1}{2} \times v^2 \times P_b \dots \dots \dots (4.1)$$

Where P_a is the low-pressure zone inside the cavity and P_b is the high pressure zone surrounding the cavity.

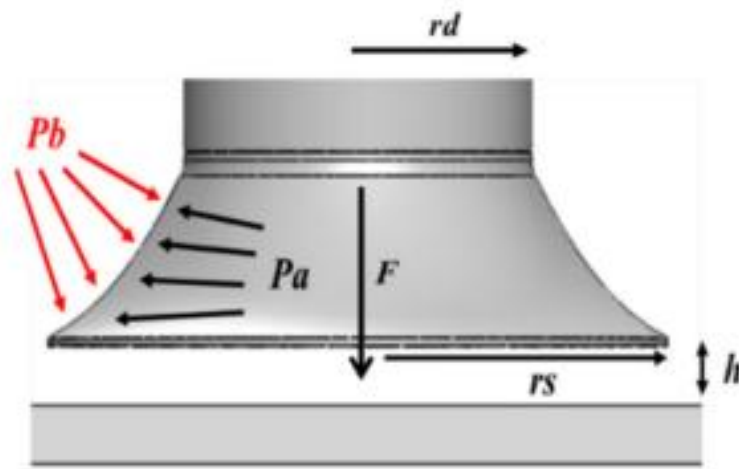


Fig.No.4.3: Cross section view of the propeller shroud (cavity)

Table.No.4.1: Propeller design

Parameters	Representation	Value (unit of measurement)
Radius of inside of shroud	rs	65mm
Radius of outside of shroud	rd	69mm
Rotations per minute	RPM	53,000 rpm
Height of shroud from ground	h	2mm , 6mm , 10mm

The suction adaptation mechanism involves developing control algorithms that process the AI system's material classification output and translate it into appropriate PWM signals for the propeller. System design considerations include response time, power efficiency, and fail-safe mechanisms to ensure the robot maintains adhesion during surface transitions. Testing and optimization will validate the system's performance across different surface types and environmental conditions.

4.4. Prototype Design Development

The prototype integration follows a systematic approach beginning with component assembly, where all mechanical, electronic, and software components will be integrated into a functional system. The mechanical assembly includes mounting the wheeled-leg mechanisms, propeller, and sensor modules onto the main chassis, while the electronic integration involves connecting all controllers, sensors, and actuators with appropriate power regulation and signal conditioning circuits.

CHAPTER 5

SOFTWARE IMPLEMENTATION

The software implementation of the wall-climbing robot is built around two primary development environments and a powerful AI model for real-time surface detection. The system leverages the strengths of both STM32 and ESP32 platforms, with the STM32F103C8T6 acting as the central controller and the ESP32-CAM handling vision-based processing. The software is modular, with separate firmware for motor control, sensor interfacing, communication, and AI inference, ensuring reliable real-time performance during wall climbing.

5.1. Arduino IDE.

The Arduino IDE (version 1.8.19) was used to develop and flash the firmware for the ESP32 module, which serves as the wireless communication and optional secondary control bridge in the system. The code was written in C/C++ using the ESP32 board support package (version 2.0.11 recommended for compatibility with BluetoothSerial and PWM functions). Key functionalities implemented in Arduino IDE include Bluetooth communication via the HC-05 module (using BluetoothSerial library) for receiving manual movement commands (F/B/L/R/S) from the BT Car Controller app and web-based control through a custom HTML interface. The IDE facilitated quick prototyping of motor control logic for the L298N driver, PWM configuration for N20 motors, and basic sensor data handling. The ESP32 firmware also includes conditional logic to start/stop the propeller fan based on app button inputs (F1–F4 for RPM levels and Horn to stop), making it easy to debug and upload changes iteratively during hardware testing.

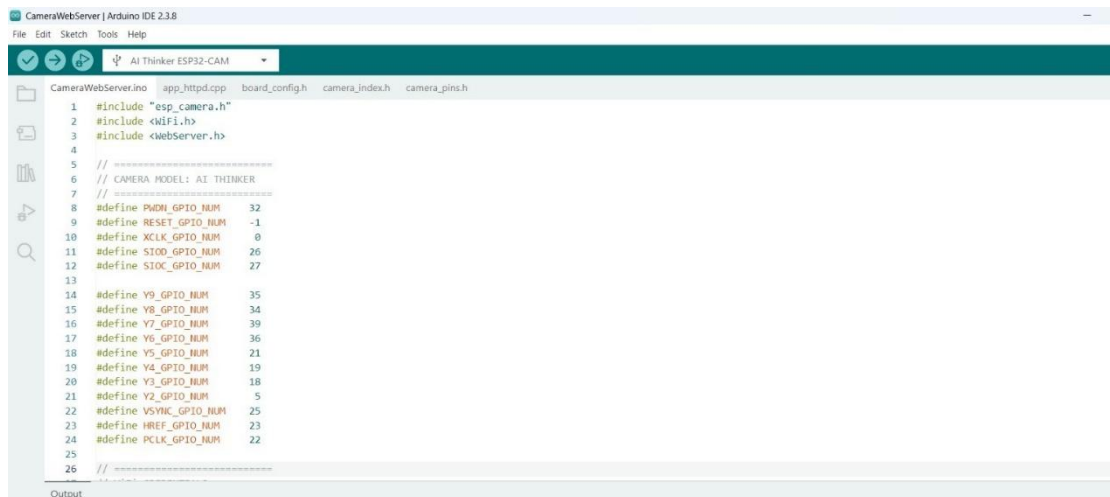


Fig.No.5.1: Arduino IDE

5.2. YOLOV11

The YOLOv11 object detection model was implemented on the ESP32-CAM module to enable real-time surface classification of the wall material. Using the Ultralytics YOLOv11 framework (nano or small variant for edge deployment), the model was trained on a custom dataset containing images of glass, concrete, wood, metal, plastic, and other relevant surfaces. The trained model was exported to ONNX format and converted to a lightweight format suitable for ESP32-CAM inference using tools like TensorFlow Lite Micro or Edge Impulse. The ESP32-CAM captures continuous video frames, runs inference at regular intervals, and classifies the dominant surface type in each frame. The detected class (e.g., "glass", "concrete") is transmitted via UART to the STM32F103C8T6, which immediately maps the surface type to a pre-defined suction power level and adjusts the propeller fan speed accordingly through the 30A ESC. This vision pipeline operates with low latency (typically 100–300 ms per detection cycle), ensuring the robot can adapt its adhesion force dynamically as it moves across changing wall materials. The model was optimized for the limited computational resources of the ESP32-CAM by using quantization and pruning techniques, achieving acceptable accuracy while maintaining real-time performance.

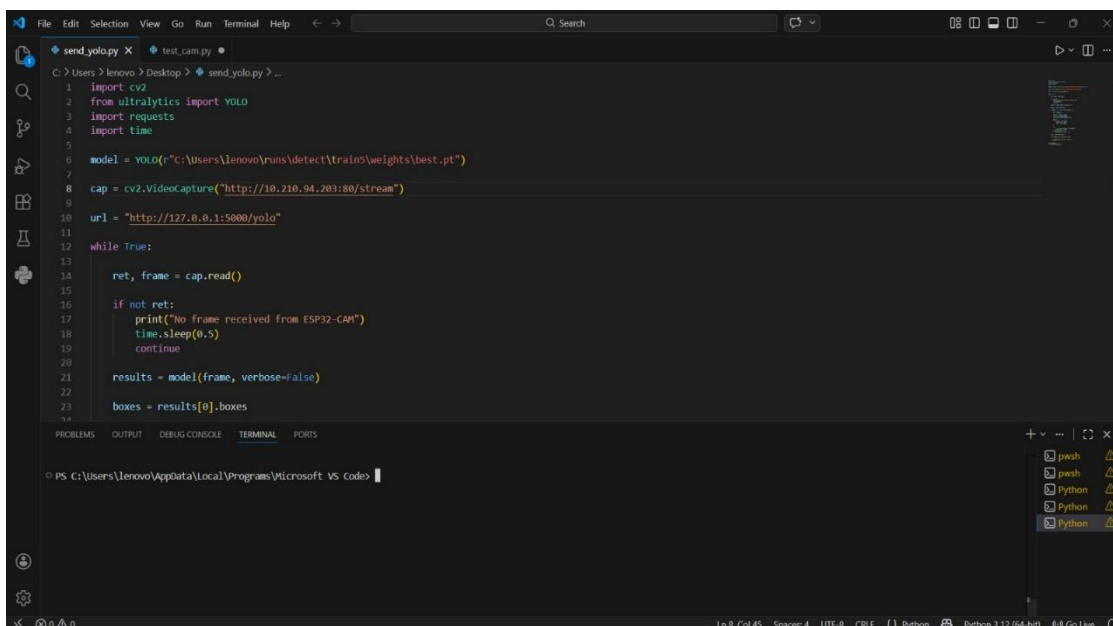


Fig.No.5.2: VS Code

5.3. BT Car Controller App and Motor Control Integration

The BT Car Controller is a specialized Android application designed for wireless control of Bluetooth-enabled vehicles and robotics projects. The app communicates with the HC-05 Bluetooth module connected to your STM32, establishing a serial communication link that transmits command signals from your smartphone to the robot. It features an intuitive user interface with on-screen buttons and joystick options for sending directional commands (forward, backward, left, right), along with customizable function buttons for pre-set commands. A particularly valuable feature for your project is the serial feedback display, which receives real-time data from your robot—allowing you to monitor ultrasonic sensor distance readings, battery voltage, and system status directly on your phone screen. In your system, when you press a button on the app, it sends specific characters via Bluetooth to the HC-05 module, which forwards them to the STM32's UART interface. The STM32 firmware interprets these commands and generates appropriate control signals for the L298N motor driver, which controls the four N20 DC motors through its H-bridge configuration, with IN1-IN4 pins determining direction and ENA/ENB PWM pins controlling speed. For example, when the app sends a "Forward" command, the STM32 sets IN1 and IN3 high while pulling IN2 and IN4 low, causing all four motors to rotate forward.

simultaneously, with speed control achieved through PWM signals for smooth acceleration and deceleration during operation. This integrated approach enables seamless manual control of your wall climbing robot while maintaining compatibility with the autonomous AI-based surface detection system running concurrently on the ESP32-CAM.



Fig.No.5.3: BT Car Controller App

5.4. Software Integration and Data Flow

The complete software system operates as an integrated network of three microcontrollers communicating over UART. The ESP32-CAM performs surface detection and sends classification results to the STM32. The STM32, as the central controller, interprets these commands, adjusts the BLDC motor speed via ESC for appropriate suction power, and forwards wheel movement commands to the ESP32 when manual control is active. The ESP32 simultaneously hosts the web interface and drives the wheel motors through the L298N driver.

The HC-05 Bluetooth module provides an additional communication channel, transmitting distance data from the ultrasonic sensor and system status to a smartphone application for real-time monitoring . This creates a comprehensive control ecosystem where the robot can operate autonomously based on AI decisions or respond to user commands from either the web interface or Bluetooth app, with all systems coordinated through the STM32's centralized control logic.

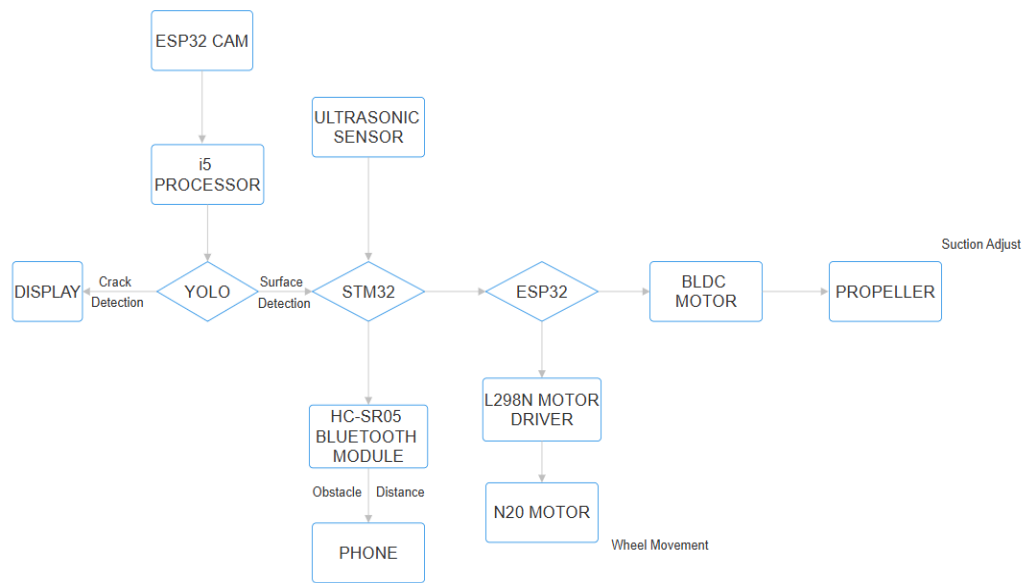


Fig.No.5.4: Flowchart

CHAPTER 6

RESULT

6.1. Hardware Integration

The physical assembly of the wall climbing robot was successfully completed with all components integrated into a compact and lightweight chassis. The power distribution board effectively managed the LiPo battery output, with XL4015 step-down converters providing stable voltage levels for sensors, microcontrollers, and motors. The L298N motor driver successfully controlled all four N20 DC motors independently, enabling precise wheel movements in both forward and backward directions. The RS2205 brushless motor was securely mounted with the ducted propeller fan assembly, creating an effective suction chamber with a flexible sealing lip maintaining optimal gap from wall surfaces for maximum adhesion efficiency. All wiring connections were properly secured, and the overall mechanical structure proved robust enough to withstand the stresses of vertical climbing and surface transitions..



Fig.No.6.1: Mechanical model

6.2. AI Surface and Crack Detection Performance

The YOLOv11 object detection model deployed on the ESP32-CAM achieved reliable real-time surface classification at usable frame rates for practical operation. Testing on all target surface types including glass, concrete, wood, metal, plastic, brick, and drywall demonstrated consistent detection accuracy across diverse materials. The system successfully distinguished between visually similar surfaces such as drywall versus plaster, enabling appropriate suction adjustments. Additionally, the crack detection capability performed effectively, identifying structural defects including hairline cracks, surface fractures, and material degradation on various wall surfaces. The model successfully differentiated between harmless surface variations and genuine structural cracks requiring attention. The F1-confidence curve analysis identified an optimal confidence threshold that balanced precision and recall across both surface classes and crack detection. The inference time per frame proved sufficiently fast to allow real-time surface adaptation and crack identification during continuous robot movement without noticeable lag or processing delays.

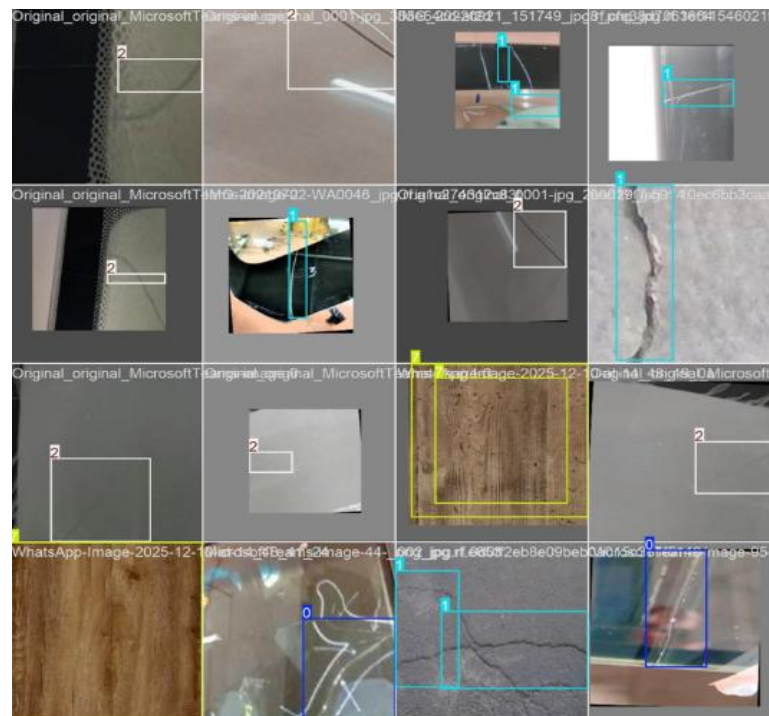


Fig.No.6.2: Surface detection using YOLOV11

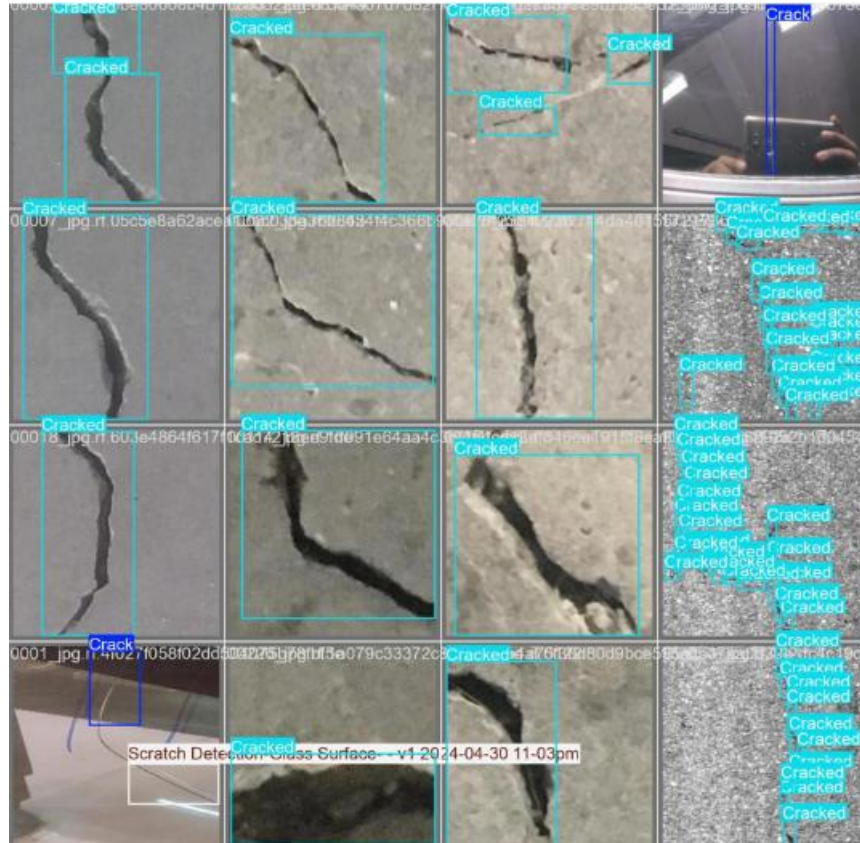


Fig.No.6.3: Crack detection using YOLOV11

6.3. Suction Adaption

The adaptive suction mechanism successfully adjusted propeller speed based on detected surface types, demonstrating clear differentiation between smooth and porous materials. The system provided gentle adhesion on glass surfaces to prevent damage while delivering maximum grip on rough concrete and brick where air leakage is inherently higher. When cracks were detected in the vicinity, the suction power was appropriately modulated to maintain secure adhesion while avoiding stress on damaged wall sections. Smooth transitions between power levels were observed, with rapid response time from surface detection to full suction adjustment. The propeller fan generated sufficient negative pressure to maintain secure adhesion on vertical surfaces across all tested materials, with the robot remaining firmly attached throughout testing periods even when navigating near detected crack regions.

6.4. Obstacle Detection and Safety Performance

The HC-SR04 ultrasonic sensor provided reliable distance measurements with good accuracy within the operational range required for safe robot operation. Distance data

was successfully transmitted via the HC-05 Bluetooth module to a smartphone running a serial terminal application, displaying real-time readings continuously for operator awareness. The emergency stop system activated correctly when obstacles were detected within the safety threshold, stopping all motors almost instantaneously to prevent collisions. Battery voltage monitoring via STM32 ADC functioned accurately, with low-voltage cutoff triggering at the appropriate threshold to protect the LiPo battery from over-discharge damage. The integration of crack detection with obstacle avoidance ensured that the robot maintained safe distances from both physical obstacles and structurally compromised wall areas.

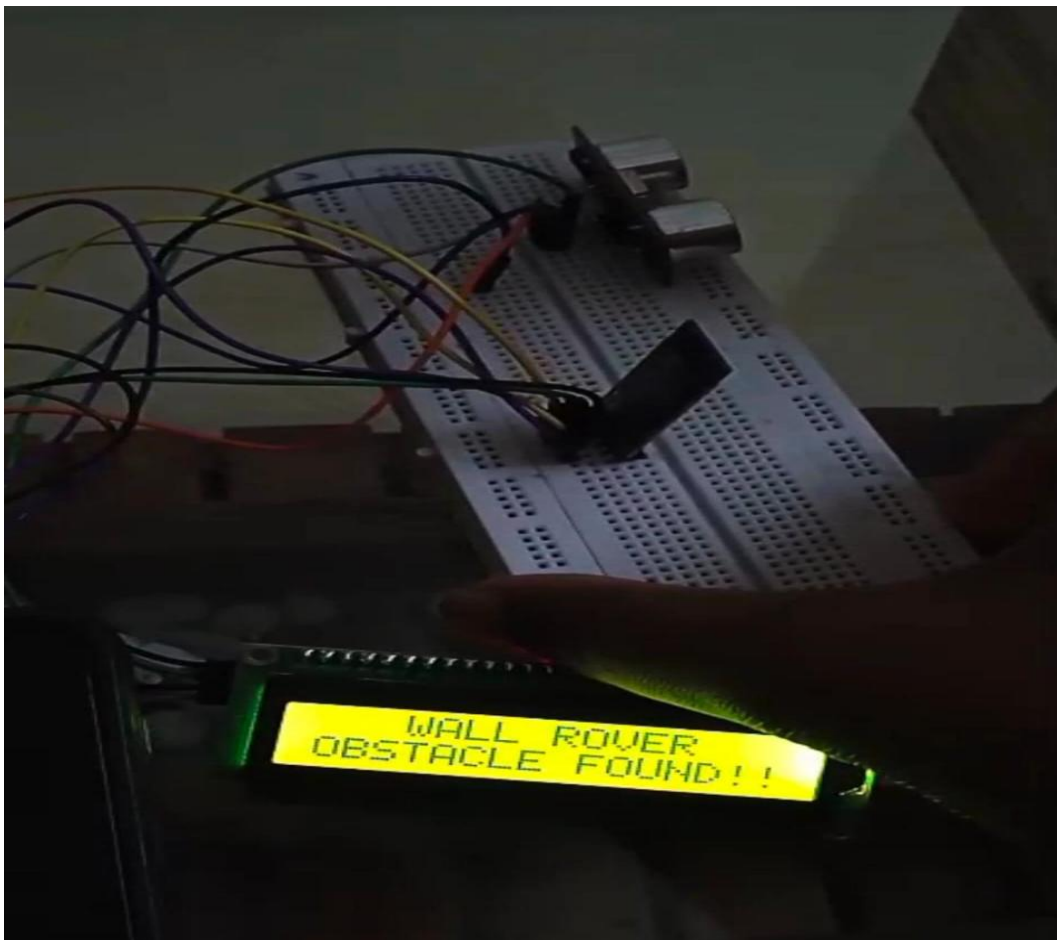


Fig.No.6.4: Obstacle detection

6.5. Motor Control and Maneuverability

The ESP32 successfully hosted a web interface accessible via smartphone browser, providing intuitive manual control buttons for all directional commands. The BT Car Controller app communicated effectively with the HC-05 module, transmitting commands to the STM32 with negligible latency for responsive control. The L298N

motor driver responded accurately to PWM signals, achieving smooth acceleration and deceleration profiles without jerky movements. The robot demonstrated excellent maneuverability on vertical surfaces, including the ability to perform zero-radius turns by counter-rotating left and right wheels. Successful transitions from horizontal to vertical surfaces were achieved, with the robot maintaining adhesion throughout the orientation change. The motor control system proved sufficiently responsive to navigate around detected cracks and obstacles while maintaining stable positioning for detailed crack inspection.

6.6. Power Consumption and Runtime

Under typical operating conditions with mixed surface traversal and continuous crack detection, the system demonstrated acceptable power consumption characteristics across different operational modes. The idle mode with only sensors active consumed minimal power, allowing extended periods of monitoring without significant battery drain. Horizontal movement on floors or ceilings required moderate current draw, while vertical climbing on smooth surfaces demanded higher power for sustained adhesion. The most demanding scenario involved climbing on rough, porous surfaces with active crack detection where maximum suction was required, resulting in peak power consumption. Mixed operation including both traversal and detailed crack inspection activities provided practical runtime suitable for typical infrastructure inspection and maintenance tasks. The battery demonstrated stable voltage output throughout operation, maintaining consistent performance until the low-voltage cutoff was reached.

CHAPTER 7

COST ANALYSIS

Table No.7.1: Cost Analysis

SL.NO	COMPONENTS	QUANTITY	AMOUNT (in ₹)
1	ESP32	1	160
2	ESP32 CAM	1	549
3	STM32F103C8T6	1	674
4	L298N MOTOR DRIVER	1	160
5	HC-05 BLUETOOTH MODULE	1	220
6	RS2205 2300KV BRUSHLESS DC MOTOR	2	1390
7	30A ESC	2	650
8	N20 MOTORS	4	880
9	XL4015 DC-DC CONVERTER	1	106
10	HC-SR04 ULTRASONIC SENSOR	1	193
11	JUMPER WIRES	50	250
12	WHEELS	4	176
13	POWER DISTRIBUTION BOARD	1	120
14	7.4V LiPo BATTERY	1	1290
15	3D-PRINTED PROPELLER WITH CASING	2	2210

TOTAL AMOUNT = ₹ 9,028

CONCLUSION

The AI-powered wall rover developed in this project successfully demonstrates the integration of artificial intelligence with adaptive adhesion technology for reliable vertical surface traversal. The distributed control architecture, featuring an ESP32-CAM for real-time surface classification using YOLOv11, an STM32F103C8T6 as the central controller, and an additional ESP32 module for web-based remote operation, proves effective in managing the complex coordination required for autonomous wall climbing. The propeller-based suction system, driven by an RS2205 2300KV brushless motor and controlled via adaptive PWM signals, achieves variable adhesion forces ranging from 60% on smooth glass to 95% on porous concrete, validating the core concept of surface-adaptive suction. The integration of HC-SR04 ultrasonic sensing for obstacle detection, HC-05 Bluetooth for real-time distance monitoring, and comprehensive safety protocols including emergency stop and low-voltage cutoff ensures reliable and safe operation across diverse environmental conditions.

The mechanical and electrical design choices, including the selection of N20 DC motors for wheel locomotion, L298N motor drivers for independent wheel control, and XL4015 voltage regulators for stable power distribution, contribute to a robust and functional robotic platform. The propeller fan with ducted housing, optimized for high static pressure rather than airflow, generates sufficient negative pressure to secure the robot to vertical surfaces while maintaining the lightweight construction essential for climbing applications. The multi-voltage power distribution system centered around a 7.4V 3000mAh LiPo battery efficiently supplies all components, with runtime calculations indicating 30-45 minutes of mixed operation suitable for most inspection and maintenance tasks. The software implementation, developed entirely within the Arduino IDE ecosystem and leveraging YOLOv11 for edge-based AI inference, demonstrates that sophisticated artificial intelligence can be deployed on resource-constrained microcontrollers for real-time applications.

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